

MS SQL Functions and Procedures

Wipro NGA
DAY - 15

FUNCTIONS

A **function** is a reusable SQL block that:

- Takes input parameters
- Performs logic
- **Returns a value or a table**
- Does NOT modify database data

Why Functions ?

- **Code reusability:** Write Once and use multiple times.
- **Cleaner logic :** INstead of programming logic
- **Maintainability:** Chnage logic at one place instead of multiple queries
- **Encapsulation:** Hide Complex business logic inside functions

Types of Functions:

- System Functions
- User Defined Functions
 1. Scalar
 2. Table valued functions (TVF)
 3. Multi-statement TVF

System Functions (Built-in)

- **1. String Functions :** UPPER, LEN
- **2. Numeric Functions:** ABS, ROUND
- **3. Date & Time Functions:** GETDATE, YEAR
- **4. Aggregate Functions:** COUNT, AVG
- **5. Conversion Functions:** CAST, CONVERT

1. Scalar Function

Returns single value

Syntax:

```
CREATE FUNCTION schema.FunctionName (@param datatype)
RETURNS datatype
AS
BEGIN
    -- logic
    RETURN value;
END;
```

Execution:

```
SELECT EmpID, FirstName, Salary,
       dbo.GetAnnualSalary(Salary) AS AnnualSalary
FROM Employees;
```

Example: Annual Salary

```
CREATE FUNCTION dbo.GetAnnualSalary (@Salary INT)
RETURNS INT
AS
BEGIN
    RETURN @Salary * 12;
END;
```

Output

EmpID	FirstName	Salary	AnnualSalary

1	John	50000	600000
2	Jane	60000	720000
3	Amit	80000	960000
4	Sara	75000	900000

2. Table-Valued Function

Syntax:

```
CREATE FUNCTION schema.FunctionName (@parameter
datatype)
RETURNS TABLE
AS
RETURN
(
    SELECT column_name
    FROM table
    WHERE condition
);
```

Execution

```
SELECT * FROM dbo.GetEmployeesBySalary(60000);
```

Example: Employees with Salary > Given Value

```
CREATE FUNCTION dbo.GetEmployeesBySalary (@MinSalary
INT)
RETURNS TABLE
AS
RETURN
(
    SELECT EmpID, FirstName, Salary
    FROM Employees
    WHERE Salary > @MinSalary
);
```

Output

EmpID	FirstName	Salary
3	Amit	80000
4	Sara	75000

limitation of Functions :

- Function are majorly used for preforming SELECT operations
- Can't perform **CRUD** operations with functions, but they can be used for :
 1. Validating Data before insert
 2. Calculating Values during INSERT/UPDATE
 3. Filtering record during select
- Transaction support is limited hence rollback and commit are not supported

Key Characteristics of Functions

- Must return a value
- Can be used in SELECT
- Functions return values and cannot modify data.
- Can not use INSERT, UPDATE, DELETE